

Visakh Padmanabhan

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LinkedIn | GitHub | Medium | Portfolio

Profile Summary

AI & Software Engineer with three years of experience in Germany delivering **end-to-end Generative AI (GenAI)** and **ML applications** across cloud platforms. Skilled in **LLM-based agentic application development**, **Retrieval-Augmented Generation (RAG)** optimization, **tool calling**, and **context engineering** in production environments. Recipient of the **Best Outgoing Student Award** (M.S., Otto von Guericke University).

Professional Experience

AI & Software Engineer

January 2026 – Present

MaindTec GmbH, Ingolstadt, Germany

- Core developer on **MaiQ**, an AI-powered project management tool for engineers, building **end-to-end agentic features** from document processing to intelligent retrieval.
- Built a **file processing service and vault management** using **Redis Streams** workers to handle PDF processing, chunking, and ingestion into **PostgreSQL** with **pgvector** for semantic retrieval.
- Conducted **11+ chunking experiments** covering document selection, question-answer generation, and retrieval evaluation using **LLM-as-judge**, and productionized the best-performing strategy, **reducing inference costs by 37%** with only a 2% drop in retrieval performance.
- Optimized **RAG** pipeline by implementing **hybrid search** combining **BM25** with embedding-based retrieval and **agentic retrieval** via **LLM tool calling** in **LangChain**, enabling page-wise and keyword-based retrieval that improved document relevance for end users.
- Implemented advanced **context engineering** techniques including chat history pruning and tool call abstraction using **agentic middlewares** in **LangChain**, enabling efficient token optimization across long conversation sessions.

Junior Data Scientist

January 2025 – August 2025

Vay Technology, Berlin, Germany

- Reduced **ETA error** by **25%** through statistical evaluation of routing performance.
- Engineered **dbt models** to monitor rebalancer missions, enabling early detection of operational issues and providing stakeholders with clear, reliable performance reports.
- Automated deployments to **AWS S3/EC2** with GitHub Actions and **CI/CD**, enabling simulations to run on higher-memory instances and improving performance at scale.

Data Scientist, Working Student

May 2022 – December 2024

Einbliq.io, Munich, Germany

- Developed **Machine Learning and predictive models** to forecast streaming device energy consumption, and built and **deployed** the Detailed Device Data Service using real-time **serverless Azure Functions**, achieving **99% coverage** of streaming data.
- Created and maintained **Grafana dashboards** to monitor streaming sessions and device capabilities, providing analysis of the Detailed Device Data Service.

Systems Engineer

October 2019 – August 2021

Infosys R&D Ltd, Thiruvananthapuram, India

- Processed and analyzed **unstructured text data** using **NLTK** and **spaCy**, applying tokenization, named entity recognition, and text preprocessing to transform raw documents into structured formats for downstream analysis.

Technical Skills

- **Programming:** Python (OOP, NumPy, Pandas, Scikit-learn), JavaScript, TypeScript, SQL
- **AI/ML & LLM Frameworks:** LangChain, LangGraph, Hugging Face, Sentence-Transformers, spaCy, NLTK, PyTorch, RAG optimization, tool calling, context engineering, information retrieval evaluation metrics
- **Software Development & APIs:** Next.js, Microservices, REST APIs, FastAPI, Flask, Streamlit
- **Databases & Vector Stores:** PostgreSQL, MongoDB, pgvector, Milvus, Redis, dbt
- **Cloud Platforms & DevOps:** AWS, Azure (Azure AI), Docker, GitHub Actions, CI/CD, MLOps
- **Tools & Workflow:** Git, JIRA, Agile/Scrum, Claude Code, Codex
- **Languages:** English (C1), German (A2)

Education

M.S. in Data and Knowledge Engineering <i>Otto von Guericke University, Magdeburg, Germany</i>	<i>October 2021 – March 2025</i> Grade: 1.4
B.Tech in Computer Science and Engineering <i>Mar Baselios College of Engineering and Technology, Thiruvananthapuram, India</i>	<i>July 2015 – August 2019</i> Grade: 1.9

Projects and Publications

- **Hybrid Legal Norm Retrieval** – Leveraging Knowledge Graphs and Sentence-BERT for retrieving Civil Code Articles (COLIEE 2024 Task 3). *Awarded Best Student Paper, JURIX 2024.* [\[Link\]](#)
 - Designed a hybrid retrieval pipeline combining **knowledge graph traversal** with **Sentence-BERT** embeddings to retrieve relevant legal norms from the Japanese Civil Code, achieving competitive results on the COLIEE 2024 benchmark.
- **LUMI: Legal Understanding and Matching through Interactive Highlighting** – Improved legal text explainability by highlighting semantically aligned spans. [\[Link\]](#)
- **Expert Agent Guided Learning** – Chatbot for guided learning using knowledge graphs built from OCR-extracted textbook content. [\[Link\]](#)
- **Slide Recommendation System** – OCR-based keyword extraction and ranking system to enhance student learning from slides. [\[Link\]](#)
- **MindMate** – Conversational AI Agent using **LangGraph**, **LangChain**, and RAG to deliver empathetic, context-aware dialogue with personalized coping strategies. [\[Personal Project Link\]](#)

Achievements

- **Best Outgoing Student Award (M.S.)** – Otto von Guericke University, Magdeburg
- **Best Student Paper Award, JURIX 2024** – Hybrid Legal Norm Retrieval: Leveraging **Knowledge Graphs** and **Sentence-BERT** for Retrieving Civil Code Articles (COLIEE 2024 Task 3)
- **All Round Performance Award** – Best Outgoing Student (2015–2019)